Digital Image Processing

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Submission deadline : 03/02 | Image Transforms | HomeWork [3]

- Students are permitted a maximum of **8 day delays** throughout the course and you are allowed to use up to 5 days of your allowed delay per homework.
- Once the allowable delay period has expired, a **20% deduction** from the total assignment score will be applied **for each additional day of delay**.
- If external resources, such as **AI tools, websites or other references**, are utilized, proper **citation and attribution** are mandatory.
- Any instance of **plagiarism or cheating** will result in a score of **0** for all parties involved. A second violation will be considered **academic dishonesty** for the entire course, with further disciplinary actions as per institutional policies.
- Please upload all your answers in a file named **HW#_[SID]_[FullName].zip** to the **CW** system.

Theoretical (55 Points)

1. Discrete Fourier Transform (24 Points)

If the Fourier transform of a function $f(x)$ is $F(u)$, determine the Fourier transform of the following functions:

a) $f(x-3)$
b) $f^2(x)$
c) $f(-x)$
d) $f(4x-3)$
e) $F(x)$

Next, consider the following 4×4 grayscale image.

33	23	13	03
32	22	12	02
31	21	11	01
30	20	10	00

f) Calculate the 2D DFT of the image.
g) Multiply the image by $(-1)^{x+y}$ and repeat the previous part.
h) How do the results relate to each other? Explain.

2. Discrete Cosine Transform (21 Points)

The definition of DCT is

$$F(u,v) = \frac{2C(u)C(v)}{\sqrt{MN}} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} \cos\left(\frac{(2i+1)u\pi}{2M}\right) \cos\left(\frac{(2j+1)v\pi}{2N}\right) f(i,j)$$

where $i, u = 0,1, \dots, M-1$, $j, v = 0,1, \dots, N-1$, and constants $C(u)$ and $C(v)$ are defined by

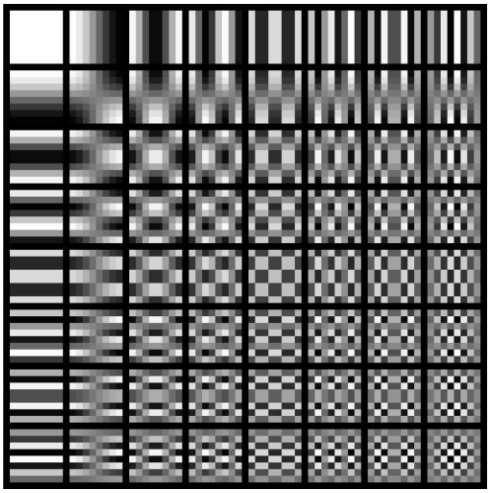
$$C(\xi) = \begin{cases} \frac{\sqrt{2}}{2}, & \xi = 0 \\ 1, & \text{otherwise} \end{cases}$$

Assume the 8×8 image $f(x,y)$ given in Fig. 1(a)

a) Determine the value of $F(0,0)$.
b) $F(0,0)$ is known as a DC coefficient. Using the computation experience from the previous part, explain what the meaning of DC coefficient is and why it is called a "DC value".
c) Fig. 1(b) displays 8×8 2-D DCT basis functions. Using the definition of DCT transform, mathematically prove that $F(0,2)$ is actually related to the pattern in the position $(0, 2)$.

10	90	160	80	10	90	160	80
20	100	150	70	20	100	150	70
30	110	140	60	30	110	140	60
40	120	130	50	40	120	130	50
50	130	120	40	50	130	120	40
60	140	110	30	60	140	110	30
70	150	100	20	70	150	100	20
80	160	90	10	80	160	90	10

(a)



(b)

Figure 1 (a) An 8x8 image given for the first part. (b) 8x8 2-D DCT basis functions.

3. Rotation (10 Points)

Fig. 2(b) was generated by:

1. Multiplying Fig. 2(a) by $(-1)^{x+y}$;
2. Computing the discrete Fourier transform;
3. Taking the complex conjugate of the transform;
4. Computing the inverse discrete Fourier transform;
5. Multiplying the real part of the result by $(-1)^{x+y}$.

Explain (mathematically) why Fig. 2(b) appears as it does.



(a) (b)

Figure 2: Rotation

Practical (45 Points)

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4. Image Analysis in Frequency Domain is Literally an Abstract Art!

The focus of this problem is mainly on what it means to represent images in the Frequency Domain, what can be inferred from Fourier Representation of an image and how image representation in spatial and frequency domain relate to each other. First, consider the images in Figure 3.

a. Specify which image(s) have Fourier transforms $F(u, v)$ with the following properties.

i. The real part of $F(u, v)$ is zero at all u, v .
ii. The imaginary part of $F(u, v)$ is zero at all u, v .
iii. $F(u, v)$ is purely real and positive for all u, v .
iv. $F(0,0) = 0$
v. $F(u, v)$ has circular symmetry

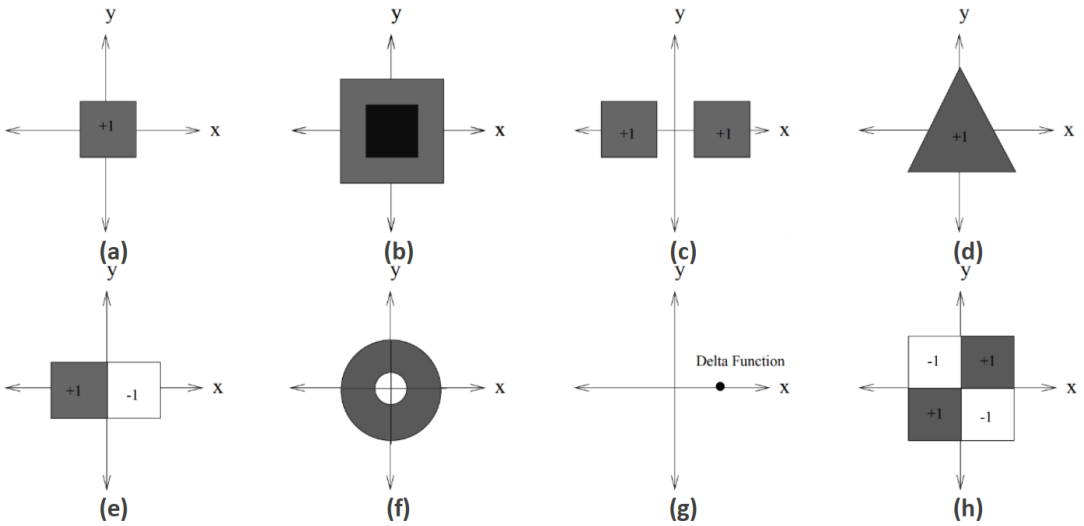


Figure 3 : Images with different structures given for the first part.

In the next part, we will get a hands-on experience with analysis of image representation in frequency domain. Consider the paintings in Figure 4.

b. Compute and display the magnitude and phase of the images. Explain which specific structure in each image yields the results you obtained. Also specify the relations between the images in the spatial domain and the frequency domain.



Good Luck!